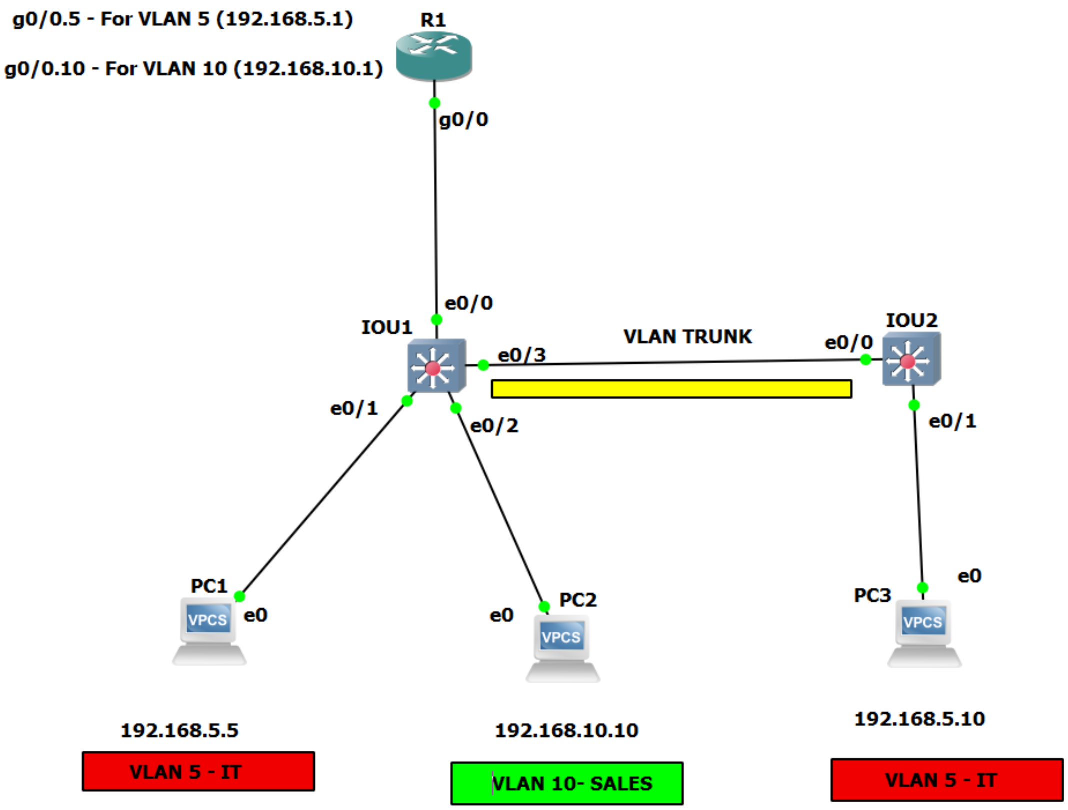
Practical No. 4

Aim: Implement Inter-VLAN Routing

Inter-VLAN Routing

Topology:



Configuration :

PC1: ip 192.168.5.5/24 192.168.5.1

Sh ip

```
PC1> ip 192.168.5.5/24 192.168.5.1
Checking for duplicate address...
PC1 : 192.168.5.5 255.255.255.0 gateway 192.168.5.1

PC1> sh ip

NAME        : PC1[1]
IP/MASK     : 192.168.5.5/24
GATEWAY     : 192.168.5.1
DNS         :
MAC         : 00:50:79:66:68:00
LPORT       : 20009
RHOST:PORT  : 127.0.0.1:20010
MTU         : 1500
```

PC2:

PC2> ip 192.168.10.10/24 192.168.10.1

PC2> sh ip

```
PC2> ip 192.168.10.10/24 192.168.10.1
Checking for duplicate address...
PC2 : 192.168.10.10 255.255.255.0 gateway 192.168.10.1

PC2> sh ip

NAME        : PC2[1]
IP/MASK     : 192.168.10.10/24
GATEWAY     : 192.168.10.1
DNS         :
MAC         : 00:50:79:66:68:01
LPORT       : 20011
RHOST:PORT  : 127.0.0.1:20012
MTU         : 1500

PC2>
```

PC3:

PC3> ip 192.168.5.10/24 192.168.5.1

PC3> sh ip

```
PC3> ip 192.168.5.10/24 192.168.5.1
Checking for duplicate address...
PC3 : 192.168.5.10 255.255.255.0 gateway 192.168.5.1

PC3> sh ip

NAME        : PC3[1]
IP/MASK     : 192.168.5.10/24
GATEWAY     : 192.168.5.1
DNS         :
MAC         : 00:50:79:66:68:02
LPORT       : 20013
RHOST:PORT  : 127.0.0.1:20014
MTU         : 1500
```

Layer2 Switch 1:

conf t

vlan 5

name IT

exit

vlan 10

name SALES

exit

end

```
IOU1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
IOU1(config)#vlan 5
IOU1(config-vlan)#name IT
IOU1(config-vlan)#exit
IOU1(config)#vlan 10
IOU1(config-vlan)#name SALES
IOU1(config-vlan)#exit
IOU1(config)#end
IOU1#
*Sep 24 14:50:26.104: %SYS-5-CONFIG_I: Configured from console by console
IOU1#
```

Layer 2 Switch 2:

conf t

vlan 5

name IT

exit

end

write memory

```
IOU2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
IOU2(config)#vlan 5
IOU2(config-vlan)#name IT
IOU2(config-vlan)#exi
IOU2(config)#end
IOU2#
*Sep 24 14:51:37.248: %SYS-5-CONFIG_I: Configured from console by console
IOU2#
```

❖ **Configuring the trunk and access interface for L2 Switch 1:**

conf t

interface ethernet0/1

switchport mode access

switchport access vlan 5

exit

interface ethernet0/2

switchport mode access

switchport access vlan 10

exit

interface ethernet0/3

switchport trunk encapsulation dot1q

switchport mode trunk

exit

interface ethernet0/0

 switchport trunk encapsulation dot1q

 switchport mode trunk

end

write memory

```
IOU1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
IOU1(config)#interface ethernet0/1
IOU1(config-if)#switchport mode access
IOU1(config-if)#switchport access vlan 5
IOU1(config-if)#exit
IOU1(config)#interface ethernet0/2
IOU1(config-if)#switchport mode access
IOU1(config-if)#switchport access vlan 10
IOU1(config-if)#exit
IOU1(config)#interface ethernet0/3
IOU1(config-if)#switchport trunk encapsulation dot1q
IOU1(config-if)#switchport mode trunk
IOU1(config-if)#exit
IOU1(config)#interface ethernet0/0
IOU1(config-if)#switchport trunk encapsulation dot1q
IOU1(config-if)#
*Sep 24 14:53:49.969: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/0, changed state to down
IOU1(config-if)#
*Sep 24 14:53:52.973: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/0, changed state to up
IOU1(config-if)#switchport mode trunk
IOU1(config-if)#
*Sep 24 14:53:58.921: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/0, changed state to down
IOU1(config-if)#
*Sep 24 14:54:01.927: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/0, changed state to up
IOU1(config-if)#exit
IOU1(config)#end
IOU1#
```

❖Configuring the trunk and access interface for L2 Switch 2:

conf t

interface ethernet0/1

 switchport mode access

 switchport access vlan 5

exit

interface ethernet0/0

 switchport trunk encapsulation dot1q

 switchport mode trunk

exit

end

write memory

```
IOU2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
IOU2(config)#interface ethernet0/1
IOU2(config-if)#switchport mode access
IOU2(config-if)#switchport access vlan 5
IOU2(config-if)#exit
IOU2(config)#interface ethernet0/0
IOU2(config-if)#switchport trunk encapsulation dot1q
IOU2(config-if)#switchport mode trunk
IOU2(config-if)#exit
IOU2(config)#end
IOU2#
*Sep 24 14:56:17.423: %SYS-5-CONFIG_I: Configured from console by console
IOU2#
```

Configuring Router R1:

```
conf t
interface Gi 0/0
 no shutdown
exit
interface Gi 0/0.5
 encapsulation dot1q 5
 ip address 192.168.5.1 255.255.255.0
 no shutdown
exit
interface Gi 0/0.10
 encapsulation dot1q 10
 ip address 192.168.10.1 255.255.255.0
 no shutdown
exit
end
write memory
```

```
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#int g0/0
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#
*Sep 24 20:27:39.395: %LINK-3-UPDOWN: Interface GigabitEthernet0/0, changed state to up
*Sep 24 20:27:40.395: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
R1(config)#interface gigabitEthernet0/0.5
R1(config-subif)#encapsulation dot1q 5
R1(config-subif)#
*Sep 24 20:28:09.631: %CDP-4-DUPLEX_MISMATCH: duplex mismatch discovered on GigabitEthernet0/0 (not half duplex), with IO
U1 Ethernet0/0 (half duplex).
R1(config-subif)#ip address 192.168.5.1 255.255.255.0
R1(config-subif)#no shutdown
R1(config-subif)#exit
R1(config)#interface gigabitEthernet0/0.10
R1(config-subif)#encapsulation dot1q 10
R1(config-subif)#ip address 192.168.10.1 255.255.255.0
R1(config-subif)#no shutdown
R1(config-subif)#
*Sep 24 20:29:09.759: %CDP-4-DUPLEX_MISMATCH: duplex mismatch discovered on GigabitEthernet0/0 (not half duplex), with IO
U1 Ethernet0/0 (half duplex).
R1(config-subif)#exit
R1(config)#
*Sep 24 20:30:09.735: %CDP-4-DUPLEX_MISMATCH: duplex mismatch discovered on GigabitEthernet0/0 (not half duplex), with IO
U1 Ethernet0/0 (half duplex).
R1(config)#
*Sep 24 20:31:09.739: %CDP-4-DUPLEX_MISMATCH: duplex mismatch discovered on GigabitEthernet0/0 (not half duplex), with IO
U1 Ethernet0/0 (half duplex).
R1(config)#
```

Testing the Network:

1) Ping PC1 to V2 member PC2 to test the connection.

PC1> ping 192.168.5.1

PC1> ping 192.168.10.1

PC1> ping 192.168.10.10

PC1> ping 192.168.5.10

```
PC1> ping 192.168.5.1

84 bytes from 192.168.5.1 icmp_seq=1 ttl=255 time=12.677 ms
84 bytes from 192.168.5.1 icmp_seq=2 ttl=255 time=15.838 ms
84 bytes from 192.168.5.1 icmp_seq=3 ttl=255 time=15.521 ms
84 bytes from 192.168.5.1 icmp_seq=4 ttl=255 time=14.943 ms
84 bytes from 192.168.5.1 icmp_seq=5 ttl=255 time=18.811 ms

PC1> ping 192.168.10.1

84 bytes from 192.168.10.1 icmp_seq=1 ttl=255 time=17.188 ms
84 bytes from 192.168.10.1 icmp_seq=2 ttl=255 time=17.197 ms
84 bytes from 192.168.10.1 icmp_seq=3 ttl=255 time=13.162 ms
84 bytes from 192.168.10.1 icmp_seq=4 ttl=255 time=8.138 ms
84 bytes from 192.168.10.1 icmp_seq=5 ttl=255 time=11.773 ms

PC1> ping 192.168.10.10

84 bytes from 192.168.10.10 icmp_seq=1 ttl=63 time=53.709 ms
84 bytes from 192.168.10.10 icmp_seq=2 ttl=63 time=37.921 ms
84 bytes from 192.168.10.10 icmp_seq=3 ttl=63 time=21.411 ms
84 bytes from 192.168.10.10 icmp_seq=4 ttl=63 time=24.964 ms
84 bytes from 192.168.10.10 icmp_seq=5 ttl=63 time=26.925 ms

PC1> ping 192.168.5.10

84 bytes from 192.168.5.10 icmp_seq=1 ttl=64 time=14.864 ms
84 bytes from 192.168.5.10 icmp_seq=2 ttl=64 time=1.489 ms
84 bytes from 192.168.5.10 icmp_seq=3 ttl=64 time=3.621 ms
84 bytes from 192.168.5.10 icmp_seq=4 ttl=64 time=3.584 ms
84 bytes from 192.168.5.10 icmp_seq=5 ttl=64 time=3.640 ms
```

2) Ping from PC2 to PCs on VLAN 5:

Ping 192.168.10.1

Ping 192.168.5.5

Ping 192.168.5.10

```
PC2> ping 192.168.10.1

84 bytes from 192.168.10.1 icmp_seq=1 ttl=255 time=28.673 ms
84 bytes from 192.168.10.1 icmp_seq=2 ttl=255 time=18.277 ms
84 bytes from 192.168.10.1 icmp_seq=3 ttl=255 time=7.271 ms
84 bytes from 192.168.10.1 icmp_seq=4 ttl=255 time=5.632 ms
84 bytes from 192.168.10.1 icmp_seq=5 ttl=255 time=12.879 ms

PC2> ping 192.168.5.5

84 bytes from 192.168.5.5 icmp_seq=1 ttl=63 time=53.739 ms
84 bytes from 192.168.5.5 icmp_seq=2 ttl=63 time=33.734 ms
84 bytes from 192.168.5.5 icmp_seq=3 ttl=63 time=30.038 ms
84 bytes from 192.168.5.5 icmp_seq=4 ttl=63 time=28.477 ms
84 bytes from 192.168.5.5 icmp_seq=5 ttl=63 time=23.806 ms

PC2> ping 192.168.5.10

84 bytes from 192.168.5.10 icmp_seq=1 ttl=63 time=54.851 ms
84 bytes from 192.168.5.10 icmp_seq=2 ttl=63 time=25.866 ms
84 bytes from 192.168.5.10 icmp_seq=3 ttl=63 time=22.361 ms
84 bytes from 192.168.5.10 icmp_seq=4 ttl=63 time=23.611 ms
84 bytes from 192.168.5.10 icmp_seq=5 ttl=63 time=20.019 ms
```

3) Similarly, ping from PC3 to the other PCs on VLAN10:

PC3> ping 192.168.5.1

PC3> ping 192.168.10.1

PC3> ping 192.168.10.10

PC3> ping 192.168.5.10

```
PC3> ping 192.168.5.1

84 bytes from 192.168.5.1 icmp_seq=1 ttl=255 time=13.282 ms
84 bytes from 192.168.5.1 icmp_seq=2 ttl=255 time=11.431 ms
84 bytes from 192.168.5.1 icmp_seq=3 ttl=255 time=14.860 ms
84 bytes from 192.168.5.1 icmp_seq=4 ttl=255 time=12.082 ms
84 bytes from 192.168.5.1 icmp_seq=5 ttl=255 time=9.762 ms

PC3> ping 192.168.10.1

84 bytes from 192.168.10.1 icmp_seq=1 ttl=255 time=5.388 ms
84 bytes from 192.168.10.1 icmp_seq=2 ttl=255 time=23.001 ms
84 bytes from 192.168.10.1 icmp_seq=3 ttl=255 time=17.595 ms
84 bytes from 192.168.10.1 icmp_seq=4 ttl=255 time=22.239 ms
84 bytes from 192.168.10.1 icmp_seq=5 ttl=255 time=32.915 ms

PC3> ping 192.168.10.10

84 bytes from 192.168.10.10 icmp_seq=1 ttl=63 time=42.871 ms
84 bytes from 192.168.10.10 icmp_seq=2 ttl=63 time=30.263 ms
84 bytes from 192.168.10.10 icmp_seq=3 ttl=63 time=24.238 ms
84 bytes from 192.168.10.10 icmp_seq=4 ttl=63 time=29.219 ms
84 bytes from 192.168.10.10 icmp_seq=5 ttl=63 time=23.432 ms

PC3> ping 192.168.5.10

192.168.5.10 icmp_seq=1 ttl=64 time=0.001 ms
192.168.5.10 icmp_seq=2 ttl=64 time=0.001 ms
192.168.5.10 icmp_seq=3 ttl=64 time=0.001 ms
192.168.5.10 icmp_seq=4 ttl=64 time=0.001 ms
192.168.5.10 icmp_seq=5 ttl=64 time=0.001 ms
```

Checking if PC1 has been correctly configured:

sh ip

```
PC1> sh ip

NAME        : PC1[1]
IP/MASK     : 192.168.5.5/24
GATEWAY     : 192.168.5.1
DNS         :
MAC         : 00:50:79:66:68:00
LPORT       : 20009
RHOST:PORT  : 127.0.0.1:20010
MTU         : 1500

PC1>
```

Checking if PC2 has been correctly configured:

sh ip

```
PC2> sh ip

NAME        : PC2[1]
IP/MASK     : 192.168.10.10/24
GATEWAY     : 192.168.10.1
DNS         :
MAC         : 00:50:79:66:68:01
LPORT       : 20011
RHOST:PORT  : 127.0.0.1:20012
MTU         : 1500
```

Checking if PC3 has been correctly configured:

sh ip

```
PC3> sh ip

NAME        : PC3[1]
IP/MASK     : 192.168.5.10/24
GATEWAY     : 192.168.5.1
DNS         :
MAC         : 00:50:79:66:68:02
LPORT       : 20013
RHOST:PORT  : 127.0.0.1:20014
MTU         : 1500
```

Checking VLAN on Layer2 Switch 1:

IOU1#show vlan bri

```
IOU1#show vlan bri

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Et1/0, Et1/1, Et1/2, Et1/3
                                                Et2/0, Et2/1, Et2/2, Et2/3
                                                Et3/0, Et3/1, Et3/2, Et3/3
5    IT                               active    Et0/1
10   SALES                            active    Et0/2
1002 fddi-default                     act/unsup
1003 token-ring-default               act/unsup
1004 fddinet-default                  act/unsup
1005 trnet-default                    act/unsup
```

Checking VLAN on Layer2 Switch 2:

IOU2#sh vlan bri

```
IOU2#show vlan bri

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Et0/2, Et0/3, Et1/0, Et1/1
                                                Et1/2, Et1/3, Et2/0, Et2/1
                                                Et2/2, Et2/3, Et3/0, Et3/1
                                                Et3/2, Et3/3
5    IT                               active    Et0/1
1002 fddi-default                     act/unsup
1003 token-ring-default               act/unsup
1004 fddinet-default                  act/unsup
1005 trnet-default                    act/unsup
```

Checking the running configuration of L2 Switch 1:

IOU1#sh running-config

```
IOU1#sh running-config
Building configuration...

Current configuration : 1739 bytes
!
! Last configuration change at 14:54:14 UTC Wed Sep 24 2025
!
version 15.1
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
service compress-config
!
hostname IOU1
!
boot-start-marker
boot-end-marker
!
!
logging discriminator EXCESS severity drops 6 msg-body drops EXCESSCOLL
logging buffered 50000
logging console discriminator EXCESS
!
no aaa new-model
no ip icmp rate-limit unreachable
!
no ip cef
!
!
no ip domain-lookup
no ipv6 cef
ipv6 multicast rpf use-bgp
spanning-tree mode pvst
spanning-tree extend system-id
!
!
!
!
!
!
vlan internal allocation policy ascending
!
ip tcp synwait-time 5
!
!
!
!
interface Ethernet0/0
```

```
!
vlan internal allocation policy ascending
!
ip tcp synwait-time 5
!
!
!
!
interface Ethernet0/0
 switchport trunk encapsulation dot1q
 switchport mode trunk
 duplex auto
!
interface Ethernet0/1
 switchport access vlan 5
 switchport mode access
 duplex auto
!
interface Ethernet0/2
 switchport access vlan 10
 switchport mode access
 duplex auto
!
interface Ethernet0/3
 switchport trunk encapsulation dot1q
 switchport mode trunk
 duplex auto
!
interface Ethernet1/0
 duplex auto
!
interface Ethernet1/1
 duplex auto
!
interface Ethernet1/2
 duplex auto
!
```

```
 no ip address
 shutdown
!
!
!
no ip http server
!
!
!
!
control-plane
!
!
line con 0
 exec-timeout 0 0
 privilege level 15
 logging synchronous
line aux 0
 exec-timeout 0 0
 privilege level 15
 logging synchronous
line vty 0 4
 login
!
end
```

Checking the running configuration of L2 Switch 2:

IOU2#sh running-config

```
IOU2#show vlan bri

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Et0/2, Et0/3, Et1/0, Et1/1
                                                Et1/2, Et1/3, Et2/0, Et2/1
                                                Et2/2, Et2/3, Et3/0, Et3/1
                                                Et3/2, Et3/3
5    IT                               active    Et0/1
1002 fddi-default                     act/unsup
1003 token-ring-default               act/unsup
1004 fddinet-default                  act/unsup
1005 trnet-default                    act/unsup
IOU2#sh running-config
Building configuration...

Current configuration : 1627 bytes
!
! Last configuration change at 14:56:17 UTC Wed Sep 24 2025
!
version 15.1
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
service compress-config
!
hostname IOU2
!
boot-start-marker
boot-end-marker
!
!
logging discriminator EXCESS severity drops 6 msg-body drops EXCESSCOLL
logging buffered 50000
logging console discriminator EXCESS
!
no aaa new-model
no ip icmp rate-limit unreachable
 --More--
```

```
!
logging discriminator EXCESS severity drops 6 msg-body drops EXCESSCOLL
logging buffered 50000
logging console discriminator EXCESS
!
no aaa new-model
no ip icmp rate-limit unreachable
!
no ip cef
!
!
no ip domain-lookup
no ipv6 cef
ipv6 multicast rpf use-bgp
spanning-tree mode pvst
spanning-tree extend system-id
!
!
!
!
!
!
vlan internal allocation policy ascending
!
ip tcp synwait-time 5
!
!
!
!
interface Ethernet0/0
 switchport trunk encapsulation dot1q
 switchport mode trunk
 duplex auto
!
interface Ethernet0/1
 switchport access vlan 5
 switchport mode access
 duplex auto
!
 --More--
```

```
 duplex auto
!
interface Ethernet3/1
 duplex auto
!
interface Ethernet3/2
 duplex auto
!
interface Ethernet3/3
 duplex auto
!
interface Vlan1
 no ip address
 shutdown
!
!
!
no ip http server
!
!
!
!
control-plane
!
!
line con 0
 exec-timeout 0 0
 privilege level 15
 logging synchronous
line aux 0
 exec-timeout 0 0
 privilege level 15
 logging synchronous
line vty 0 4
 login
!
end
```

❖ **The interfaces the have been set to the trunk mode in L2 Switch 1:**

IOU1#sh int trunk

```
IOU1#sh int trunk

Port        Mode             Encapsulation  Status        Native vlan
Et0/0       on               802.1q         trunking      1
Et0/3       on               802.1q         trunking      1

Port        Vlans allowed on trunk
Et0/0       1-4094
Et0/3       1-4094

Port        Vlans allowed and active in management domain
Et0/0       1,5,10
Et0/3       1,5,10

Port        Vlans in spanning tree forwarding state and not pruned
Et0/0       1,5,10
Et0/3       1,5,10
IOU1#
```

The interfaces the have been set to the trunk mode in L2 Switch 2:

IOU2#sh int trunk

```
IOU2#sh int trunk

Port        Mode             Encapsulation  Status        Native vlan
Et0/0       on               802.1q         trunking      1

Port        Vlans allowed on trunk
Et0/0       1-4094

Port        Vlans allowed and active in management domain
Et0/0       1,5

Port        Vlans in spanning tree forwarding state and not pruned
Et0/0       1,5
```

The overall running configuration of Router 1 (R1):

R1#sh running-config

```
R1#sh running-config
Building configuration...

Current configuration : 1138 bytes
!
! Last configuration change at 20:42:35 UTC Wed Sep 24 2025
!
version 15.2
service timestamps debug datetime msec
service timestamps log datetime msec
!
hostname R1
!
boot-start-marker
boot-end-marker
!
!
!
no aaa new-model
no ip icmp rate-limit unreachable
ip cef
!
!
!
!
!
!
no ip domain lookup
no ipv6 cef
!
!
multilink bundle-name authenticated
!
!
!
!
!
!
!
!
!
ip tcp synwait-time 5
```

```
!
interface Ethernet0/0
 no ip address
 shutdown
 duplex auto
!
interface GigabitEthernet0/0
 no ip address
 media-type gbic
 speed 1000
 duplex full
 negotiation auto
!
interface GigabitEthernet0/0.5
 encapsulation dot1Q 5
 ip address 192.168.5.1 255.255.255.0
!
interface GigabitEthernet0/0.10
 encapsulation dot1Q 10
 ip address 192.168.10.1 255.255.255.0
!
interface GigabitEthernet1/0
 no ip address
 shutdown
 negotiation auto
!
ip forward-protocol nd
!
!
no ip http server
no ip http secure-server
!
!
!
!
control-plane
!
!
line con 0
 exec-timeout 0 0
 privilege level 15
 logging synchronous
 stopbits 1
line aux 0
 exec-timeout 0 0
 privilege level 15
 logging synchronous
 stopbits 1
line vty 0 4
 login
!
!
```